

Human-Data Interaction, Exploration & Visualization in the AI Era: Challenges & Opportunities

Jean-Daniel Fekete¹ • Yifan Hu² • Dominik Moritz³ • Arnab Nandi⁴ • Senjuti Basu Roy⁵ • Eugene Wu⁶ • Nikos Bikakis⁷ • George Papastefanatos⁸ • Panos K. Chrysanthis⁹ • Guoliang Li¹⁰ • Lingyun Yu¹¹

¹Univ. Paris-Saclay & Inria • ²Northeastern • ³CMU & Apple • ⁴Ohio State • ⁵NJIT • ⁶Columbia • ⁷Hellenic Med. Univ. & Archimedes/Athena RC • ⁸Athena RC • ⁹Pittsburgh • ¹⁰Tsinghua • ¹¹XJTLU



Problem

AI-era data analysis now spans **large-scale, heterogeneous, multimodal, and predominantly unstructured** data — text, images, audio, video, and visually rich documents — while foundation models (LLMs, VLMs) inject **uncertainty** into every step.

This breaks the schema-driven query-then-explore workflows systems were built on: data is weakly indexed and costly to inspect.

Users must iteratively explore, reformulate, and verify results under uncertainty — often without knowing what the data contains or which questions can be meaningfully asked.

Motivation

- **Perceptual latency is a hard constraint.** Delays of even a few seconds disrupt reasoning, bias exploration, and reduce the effectiveness of human-AI collaboration.
- **Foundation models are double-edged.** LLMs and VLMs add non-determinism, error propagation, bias, and verification cost — reinforcing the need to keep humans meaningfully in the loop.

DATA SYSTEMS TODAY

Tuned for **order-of-second** responses.

vs.

HUMAN COGNITION

Needs **order-of-millisecond** feedback.

Meeting these demands requires interface-, perception-, and context-aware systems, with explicit support for oversight, provenance, and trust calibration.

Method

REFRAME

Stop optimizing data management, AI, interfaces, and visualization in isolation — **co-design them as one end-to-end, human-centered pipeline** for iterative, guidance-driven, mixed-initiative exploration.

Four coupled perspectives trace the loop: how interfaces shape systems, millisecond querying, scalable analytics, and AI-guided exploration with generative visualization.

1

Interface-System Co-Design

Sec. 2

- Treat visualization as a **SQL-workload generator**: interactions become queries, so faster execution improves responsiveness.
- But P95 latency and throughput don't equal fluid UX — the interface's *own* constraints motivate new data-management problems.

System design can also **inspire** new interfaces, not merely react.

2

Queries at the Speed of Thought

Sec. 3

ML shifted from model-centric to **data-centric** — the bottleneck is now understanding data, not tweaking models.

Engineers slice and dice millions of records to grasp subtle patterns, and visualization allocates scarce attention — but only if response times match human cognition. Data systems tuned for order-of-second responses distort exploration; approximation, indexing, and prefetching buy interactivity at a cost in complexity.

3

Scalable Visual Analytics

Sec. 4

~2% of IEEE VIS papers use a scalability keyword

Progressive visual analytics and approximate query processing help, and stacks like Mosaic abstract low-level optimizations — but a single consensual measure of “scalable visualization” is still missing.

4

AI-Guided Exploration & Generative Vis

Sec. 5-7

- Unstructured, metadata-poor data demands **ad-hoc, continuous query reformulation**; LLMs extract structure but add non-determinism, cost, and bias.
- A trained **GNN** optimizes arbitrary differentiable layout objectives on unseen graphs; a **GAN** optimizes non-smooth objectives from only pairwise “which is better” comparisons.

This opens a path to modeling **human aesthetic preference** directly — visualization as an active, narrative, adaptive component.

Headline Numbers

~2%

of IEEE VIS papers use a scalability keyword

far below sibling data-science domains

ms

latency target vs. order-of-second systems

11

authors across 10+ institutions

6

coordinated section perspectives

4

surveyed research literatures

Key Results

DIMENSION	LEGACY VIEW	AI-ERA DEMAND
Latency	order-of-second	order-of-millisecond
Data	structured tables	multimodal, unstructured
Visualization	static output	active, generative
Human role	operator	in-the-loop overseer

Only ~2% of IEEE VIS papers engage scalability — a fundamental barrier vs. sibling data-science domains.

SO WHAT → *Deep learning even reshapes visualization itself: GNN/GAN models learn layout from data.*

Takeaway

AI-driven ecosystems are an **inflection point**: interactive systems can no longer be built by independently optimizing data management, AI, and interaction — they must be **co-designed end-to-end**, with visualization shifting from passive display to an active, human-guided role.

Databases

Foundation Models

InfoVis

HCI

Comp. Graphics

Cognitive Science

Scan to Read



Paper